

Final Submission: Knowledge Graph for Lung Cancer Diagnosis and Treatment

1. Documentation

Introduction

This project focuses on building a comprehensive knowledge graph for Lung Cancer, designed to provide a structured representation of the relationships between symptoms, treatments, diagnostic tests, medications, and associated factors. The knowledge graph aims to support clinical decision-making and facilitate research in the healthcare domain.

Scope

The knowledge graph covers a wide range of entities related to lung cancer, including:

Diseases:

Lung Cancer and its subtypes, such as Non-Small Cell Lung Cancer and Metastatic Lung Cancer.

Symptoms:

Common symptoms like Cough, Chest Pain, Shortness of Breath, and more.

Treatments:

Methods such as Chemotherapy, Radiation Therapy, and Surgery.

Medications:

Drugs like Cisplatin, Atezolizumab, and Docetaxel.

Diagnostic Tests:

Procedures like Biopsy, CT Scan, MRI, and PET Scan.

Lifestyle Changes:

Preventive measures such as Quit Smoking, Healthy Diet, and Regular Exercise.

Support Resources: Services like Cancer Care Services and Lung Cancer Support Groups.

Data Collection

To ensure accuracy and relevance, data was sourced from trusted platforms:

- MedlinePlus for diseases and treatments.
- SNOMED CT for standardized clinical terms.
- DrugBank for medication details.

CDC/WHO for global health statistics.

The data underwent preprocessing steps, including cleaning duplicates, resolving inconsistencies in terminology, and standardizing entity names for integration into the knowledge graph.

Schema Design

Entities: Core components include Disease, Symptom, Treatment, Medication, Diagnostic Test, and more.

Relationships: Defined connections, such as Disease-HasSymptom, Disease-CanBeTreatedWith, and Treatment-UsesMedication.

Attributes: Each entity has relevant attributes, such as Effectiveness for treatments and Sensitivity for diagnostic tests.

Knowledge Graph Creation

The knowledge graph was built using WebProtégé, an ontology editor. The structured data was transformed into an OWL (Web Ontology Language) file, enabling compatibility with tools for querying and visualization.

2. Working Knowledge Graph

Description:

The knowledge graph represents lung cancer's relationships with symptoms, treatments, diagnostic tests, and associated factors. It serves as a functional resource to explore these connections and retrieve insights.

OWL File:

The OWL file has been exported, making the knowledge graph accessible for querying and further development.

3. Queries

a. What are the symptoms of Lung Cancer?

```
SELECT ?symptom
WHERE {
  ?disease :hasSymptom ?symptom .
  ?disease rdf:label "Lung Cancer" .
}
```

b. What medications are used in Chemotherapy?

```
SELECT ?medication
WHERE {
  ?treatment :usesMedication ?medication .
  ?treatment rdf:label "Chemotherapy" .
}
```

c. What lifestyle changes can help reduce the risk of Lung Cancer?

```
SELECT ?lifestyleChange
WHERE {
  ?lifestyleChange :reducesRisk ?disease .
}
```

```

    ?disease rdf:label "Lung Cancer" .
}

```

4. Visualization

The graph visualization offers a clear and intuitive representation of lung cancer's connections with related entities. Below is a visual snapshot of the knowledge graph:

